

MOROZOVSK, Russian FederationWMO: **345450**Lat: **48.350N**Lon: **41.867E**Elev: **294**StdP: **14.54**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.7	-1.3	-12.4	2.8	-5.9	-7.0	3.8	-0.5	27.4	26.8	24.8	27.3	4.8	80	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.6	96.5	66.4	92.8	65.9	89.2	65.1	69.9	85.0	68.6	84.0	67.4	82.9	10.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.6	95.5	74.3	63.9	90.0	73.1	62.4	85.2	72.4	34.1	84.7	33.0	83.7	31.9	82.8	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.4	20.9	18.1		-12.5	99.4	7.4	3.8	-17.9	102.1	-22.2	104.3	-26.3	106.4	-31.7	109.2	
				-12.9	74.3	7.3	4.0	-18.2	77.2	-22.5	79.6	-26.6	81.8	-32.0	84.8	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.4	22.7	24.2	34.6	49.2	61.3	69.3	74.5	72.4	60.9	47.9	35.5	26.7	
	DBStd	20.48	11.65	11.68	8.27	7.68	8.00	6.92	6.27	7.33	7.68	8.99	9.53	10.79	
	HDD50	3475	846	721	479	104	9	0	0	0	8	147	439	722	
	HDD65	6864	1311	1141	942	474	169	34	3	18	167	534	884	1187	
	CDD50	2896	0	0	2	81	359	580	760	693	335	81	5	0	
	CDD65	809	0	0	0	0	54	164	299	246	43	3	0	0	
	CDH74	9672	0	0	0	34	681	1843	3539	3067	476	32	0	0	
	CDH80	4270	0	0	0	4	204	754	1702	1480	120	6	0	0	
Wind	WSAvg	8.2	9.1	9.9	9.9	9.1	7.6	7.2	6.6	6.7	7.2	7.2	8.4	9.1	
Precipitation	PrecAvg	17.3	1.5	1.2	1.3	1.0	1.8	1.7	1.7	1.1	1.7	1.4	1.3	1.7	
	PrecMax	22.9	2.8	3.3	2.9	2.0	5.7	4.3	5.8	2.8	4.1	3.1	3.4	3.0	
	PrecMin	10.7	0.7	0.3	0.2	0.2	0.0	0.4	0.3	0.1	0.1	0.1	0.2	0.6	
	PrecStd	3.2	0.6	0.7	0.7	0.6	1.3	1.0	1.3	0.7	1.1	0.9	0.7	0.7	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.7	48.9	61.0	78.8	90.8	96.3	100.1	101.9	87.3	78.4	60.4	47.9	
		MCWB	42.6	44.3	49.2	57.9	64.6	65.8	67.0	66.6	64.0	61.2	53.2	45.9	
	2%	DB	41.2	42.8	54.7	71.8	85.2	91.8	96.8	96.7	83.5	70.5	54.9	44.9	
		MCWB	39.7	39.9	45.6	55.9	62.1	65.8	66.7	66.3	61.5	57.4	50.1	43.4	
	5%	DB	37.9	39.5	50.4	67.5	81.4	88.1	93.1	92.5	79.7	65.4	50.8	42.0	
		MCWB	36.5	37.6	43.6	53.3	61.5	65.4	66.1	65.6	60.9	55.9	47.6	40.4	
	10%	DB	35.5	36.8	46.5	63.7	77.3	84.2	89.2	88.3	75.5	61.0	47.9	39.1	
		MCWB	34.5	35.3	41.6	51.2	60.4	64.2	65.5	64.6	59.4	54.0	45.6	37.9	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.2	44.6	50.8	60.5	67.5	71.5	71.9	71.6	68.5	63.7	54.0	46.4
			MCDB	45.1	47.9	58.0	74.8	83.7	85.9	87.7	85.3	81.8	74.6	58.6	47.0
2%		WB	39.6	40.4	47.1	57.3	65.2	69.1	70.0	69.3	65.5	59.8	51.0	43.4	
		MCDB	41.2	42.2	52.5	69.0	79.5	83.5	84.8	85.5	76.6	66.2	54.3	44.8	
5%		WB	36.7	37.8	44.3	55.2	63.3	67.5	68.7	67.8	63.3	57.1	48.1	40.6	
		MCDB	37.6	39.2	49.2	64.7	76.6	81.7	84.1	84.2	73.7	63.5	50.4	41.8	
10%		WB	34.6	35.5	41.7	53.0	61.6	65.9	67.5	66.4	61.2	54.7	45.7	37.9	
		MCDB	35.3	36.6	46.1	61.3	74.3	79.9	83.3	83.5	70.9	60.5	47.8	39.1	
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.9	13.3	19.9	21.6	21.8	22.6	24.3	21.7	16.7	11.2	9.1	
		MCDBR	7.8	11.3	20.8	25.8	27.3	27.3	27.4	29.3	26.9	23.4	15.0	9.3	
		MCWBR	7.2	9.1	13.1	12.5	10.8	9.5	8.2	9.8	11.4	12.9	10.8	8.3	
	5% WB	MCDBR	7.8	10.5	18.4	22.0	23.2	22.4	22.6	23.6	22.1	19.9	13.5	9.1	
		MCWBR	7.4	8.7	12.6	11.6	10.4	9.4	8.4	9.9	11.2	12.2	10.4	8.4	
Clear-Sky Solar Irradiance	taub	0.333	0.366	0.398	0.449	0.417	0.412	0.438	0.438	0.410	0.377	0.349	0.336		
	taud	2.152	2.070	2.130	2.067	2.253	2.302	2.235	2.214	2.261	2.305	2.287	2.192		
	Ebn at Noon	220	239	254	253	269	270	261	254	249	237	216	202		
	Edh at Noon	27	37	41	48	42	40	42	41	36	29	24	23		
All-Sky Solar Radiation	RadAvg	310	596	959	1421	1878	2033	1966	1791	1273	726	340	212		
	RadStd	73	117	124	139	178	172	108	121	122	99	64	32		